

Abdulrooheem Olamide Usman

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Summary

AI-focused Mechatronics Engineer building intelligent systems across healthcare and fintech, with a focus on real-world ML deployment, system design, and data-driven decision making. Experienced in translating data into practical, deployable solutions.

Links

GitHub: <https://github.com/usman-abdul>

Featured Projects

Resolve AI - Behavioral Intelligence System

- Built a full-stack behavioral intelligence system using **AWS Lambda, API Gateway, and DynamoDB (single-table design)**.
- Designed a **deterministic decision engine** that analyzes user behavior (completion rate, session duration, consistency) to generate structured recommendations.
- Integrated **Amazon Bedrock (Claude 3 Haiku)** to convert system outputs into natural language with fallback handling.
- Implemented authentication using **AWS Cognito (JWT)** and developed a frontend with **React + TypeScript**.
- Focused on **scalable architecture, modular design, and separation of logic from AI generation**.

DeepFraud AI - Financial Fraud Detection System

- Developed an anomaly detection system for financial transactions using **Isolation Forest** to identify suspicious **behavioral patterns**.
- Built a real-time detection pipeline with a **Streamlit dashboard** for monitoring anomaly scores and flagged transactions.
- Designed a **scoring and alert system** that prioritizes consistency with model outputs while providing interpretable explanations.
- Explored system extensions for **sequence-based detection** and advanced models (**Autoencoder, LSTM**) as part of future improvements.
- Focused on building a **lightweight, explainable system** suitable for **real-time fraud monitoring** in fintech systems.

Blood Pressure Prediction System

- Developed a machine learning model to predict **hypertension risk** using **Logistic Regression** on structured health data.
- Identified a critical **dataset bias** where predictions were dominated by a single feature, leading to misleadingly high accuracy.
- Improved **model generalization** by introducing **synthetic data** to create more realistic feature relationships.
- Evaluated **model performance** using accuracy and classification metrics, emphasizing the trade-off between accuracy and real-world reliability.
- Focused on building **interpretable models** and understanding **data quality** in healthcare applications.
- Achieved **~90% accuracy** after improving data quality and model generalization.

Experience

National Center for Artificial Intelligence & Robotics (NCAIR) | Abuja

Internship Student | 03/2024 - 08/2024

- Developed and evaluated machine learning models (logistic regression, random forest) for health data classification, improving predictive accuracy and system reliability.
- Created a license plate recognition system using Python and OpenCV with real datasets capable of processing static images with improved detection accuracy.
- Designed dashboards and improved UI/UX flows using Figma for data-driven applications.
- Developed IoT systems integrating sensors with embedded controllers.

Skills

Technical

- **Programming:** Python, SQL
- **AI / Data:** Scikit-learn, TensorFlow, Pandas, NumPy
- **Embedded & IoT:** ESP32, Arduino, Sensors & Actuators
- **Design Tool:** Figma

Professional

- System Design • Team Collaboration
- Technical Documentation • Rapid Prototyping

Education

Nile University of Nigeria | Abuja
Bachelor of Engineering (B.Eng) in Mechatronics Engineering | 10/2025
GPA: 4.31 / 5.0

Languages: English, Yoruba
